VNR Vignana Jyothi Institute of Engineering

Electronics and Communication Engineering | Course Code: EC202

Signals and Systems - Comprehensive Syllabus Lecture Notes

Syllabus Focus: EC202 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Signals and Systems. Includes detailed unit-b

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Signal Classifications & System Properties

" Syllabus Scope: Continuous vs discrete time, energy vs power signals, linearity, time-invariance, causa

Unit 2: LTI Systems & Convolution

" Syllabus Scope: Convolution integral, convolution sum, impulse response, step response, cascade and par

EC202: Signals and Systems - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Fourier Analysis of Continuous-Time Signals

" Continuous-Time Fourier Series CTFS, Dirichlet conditions, Continuous-Time Fourier Transform CTFT, duality,

Unit 4: Laplace Transform & System Stability

" Region of Convergence ROC properties, unilateral and bilateral Laplace transforms, transfer functions, pole-

Unit 5: Z-Transform & Discrete-Time Analysis

" Z-Transform definition, ROC properties, inverse Z-Transform partial fractions, residue, stability and causal

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Convolve two continuous-time rectangular pulses xt = ut - ut-2 and ht = ut - ut-1 analytically.
2. State and prove the Nyquist Sampling Theorem, explaining aliasing distortion and anti-aliasing filter
3. Find the Laplace transform and ROC of signal xt = e^-3tut + e^-2tu-t.

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Alan V. Oppenheim, Alan S. Willsky - Signals and Systems Pearson 2nd Ed
* Simon Haykin, Barry Van Veen - Signals and Systems Wiley 2nd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students